SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	NoBugPls
Project / Product Name	AiversePulse Enterprise
Final Live Deployment URL	https://sdc.olgtx.com
Codebase Repository URL	https://github.com/yushan1101/SDC_Hackathon_Submission_NoBugPls.git
Name (Leader)	TANG YU SHAN
Email & phone (Leader)	tangys2005@gmail.com & +60175908412

2. Project Description

[Maximum 250 words]

AiversePulse Enterprise helps individuals and organisations prepare for changes in work caused by AI. It connects skills assessment, career exploration, personalised learning and workforce planning in one application.

Individuals begin with a career questionnaire, resume upload or selected target occupation. The system matches their profile against 2,938 ESCO occupations, identifies skill gaps under assistive and agentic AI scenarios, and creates a personalised 30/60/90-day reskilling plan constrained by their available learning hours. Plans include learning references, practical activities and observable deliverables. Adaptive mock interviews provide additional practice and feedback.

For HR teams, the platform supports team-data imports, workforce skill-gap heatmaps, shared training-plan drafts, job openings, résumé screening and blind candidate review. Human reviewers retain responsibility for plan approval and hiring decisions.

LangGraph coordinates specialist agents, while FastMCP provides access to occupational knowledge, public AI-trend signals and forecasting tools. Optional generative AI support personalised explanations and learning content; deterministic checks validate structured outputs and constrain recommendations.

The project supports SDG 4, particularly Target 4.4 on employment-relevant skills, and SDG 8 on decent work. Its current contribution is an evidence-linked prototype for reskilling decisions. Improvements in employment, wages or completed learning have not yet been demonstrated through a longitudinal pilot.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

AiversePulse Enterprise is a web application that integrates a browser-based frontend, a FastAPI backend, a multi-agent analysis workflow, GenAI APIs, and persistent storage into a unified career-planning and HR support platform. The frontend is hosted on Replit, while the backend API runs on a dual-node VPS. Both the learner-facing experience and the HR console use the same backend services, with separate workflows for career planning and resume review.

The frontend is built using HTML, CSS, and JavaScript. It supports profile creation through a skills quiz, resume upload, or manual skill entry; role matching; learning plans; interview practice; and HR review tools. For longer analyses, the browser receives streamed progress updates from the backend. Resume files are processed in memory, with text extraction and document-safety checks performed before the extracted content is used.

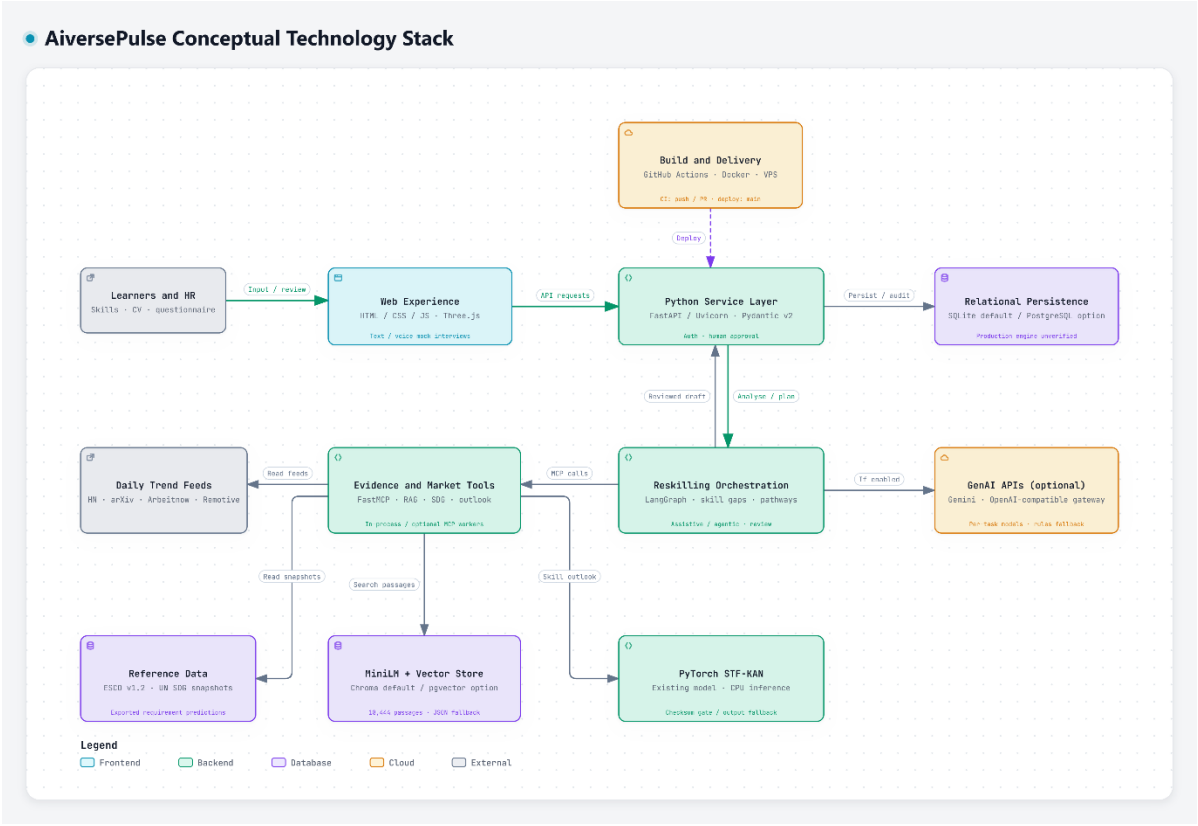
The backend coordinates analysis through LangGraph. A supervisor routes requests, while the profiler, knowledge retrieval, and market analysis steps can run in parallel. The workflow then analyzes skill gaps, drafts a 30/60/90-day pathway, generates a cited explanation, and runs a reviewer that checks evidence, plan workload, and other constraints. Plans remain drafts until approved by a designated human. Resume screening follows a separate workflow that extracts skills, evaluates configured criteria, and presents findings for human review.

GenAI APIs provide an optional service layer for tasks such as explanations, coaching, resume-skill proposals, and interview responses. The application can route these tasks to configured Gemini or OpenAI-compatible APIs. The core workflows also include deterministic behavior and can operate without a GenAI API key, allowing the basic demo and career-planning flow to continue when a provider is unavailable.

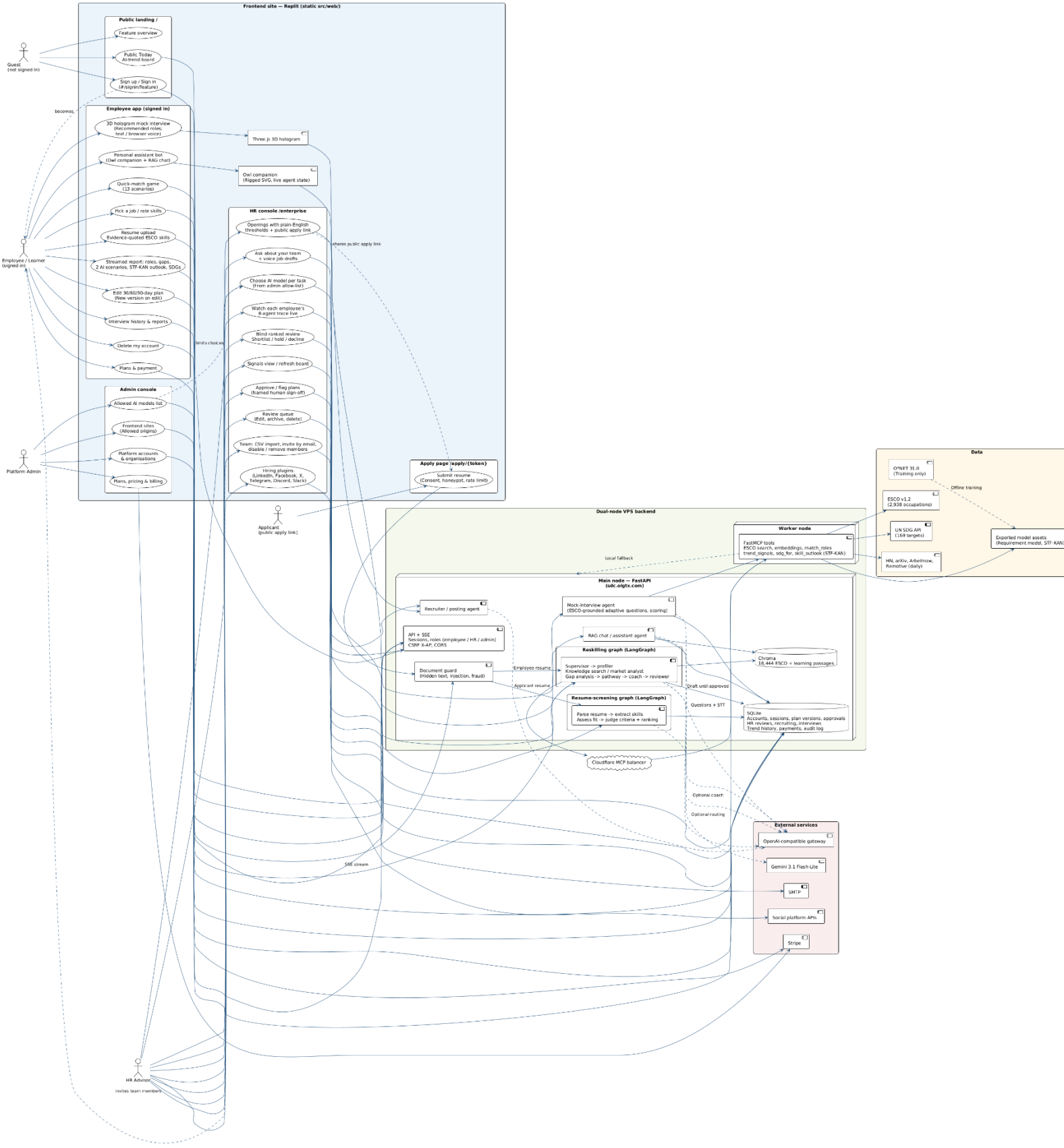
Knowledge and market evidence come from several sources. ESCO occupation and skill data, a prebuilt vector index of ESCO and learning-resource passages, and a snapshot of UN SDG goals and targets support role matching, retrieval, and SDG mapping. Daily trend signals are gathered from public feeds and stored with dates and source information. The application serves precomputed STF-KAN skill-outlook results. O*NET is used to train and evaluate models, while the live application uses exported model outputs rather than reading the training

data directly. SDG associations are semantic suggestions, with a documented baseline used when no confident role match is available.

SQLite stores application records such as accounts, plans, plan versions, approvals, and audit events. Chroma stores vector representations used for semantic retrieval. Model and data snapshots are maintained as files shipped with the application. The frontend communicates with the FastAPI backend over API endpoints, while the backend coordinates workflow execution, data access, and external GenAI API calls.



AiversePulse Enterprise — Roles, Features and Architecture



3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

AiversePulse builds on the Phase 1 prototype, which already described ESCO-based role matching, multi-agent reskilling plans, resume screening, RAG chat, a Three.js mock interviewer and a Replit frontend connected to a dual-node backend.

The current implementation extends that foundation with more detailed learning journeys, streamed plan content and stronger organisation-level workforce workflows. Learning plans include practical instructions, exercises, deliverables, knowledge checks and completion criteria. Generated content remains connected to structured skill gaps, retrieved resources and available learning time.

HR teams can import workforce datasets, compare gaps by department, role, dataset or group, and prepare separate team-plan drafts for review. Review-queue controls support candidate updates, filtering, archiving and bulk actions. Plan-status changes retain version history and require explicit confirmation.

PostgreSQL and pgvector support shared operational records and knowledge retrieval across application nodes. Configurable model assignments and usage controls support different AI tasks and organisation requirements.

A key implementation challenge was combining personalised AI-generated content with predictable planning constraints. The application addresses this through structured schemas, citation-ID checks, phase-hour limits, bounded repairs and fallback content. Human approval remains separate from AI generation.

The final submission should link sprint-specific improvements to repository commits and demonstrations. Features already documented in Phase 1 are the project foundation and should not be presented as newly created during Phase 2.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Our Phase 1 proposal defined evidence compliance as the percentage of plan activities citing both a dated source and an SDG target. The Phase 2 targets were at least 95% compliance on the first draft, with 100% of activities presented for approval either verified or explicitly flagged. The prototype checks the retrieved citation IDs, documented skill gaps, positive activity hours and learning-time limits. It allows up to 2 targeted repair attempts and flags unresolved validation errors. However, these checks do not fully verify source dates and activity-level SDG targets. The aggregate compliance evaluation has not yet been completed which makes the achievement of both targets remains unverified. For the secondary rising-skill KPI, the saved historical evaluation recorded 1.389 × lift (approximately 1.39) against a 2.09% base rate. This supports the previously reported historical result but does not establish performance after a new data refresh. These findings demonstrate implemented planning controls and historical ranking performance in support of SDG Targets 8.5 and 4.4. They do not demonstrate improved employment or completed learning outcomes.
Verification Evidence Method *How was this data verified?*	We verified the prototype's evidence and forecasting claims against the codebase, exported model cards, and held-out evaluation logs. The Reviewer checks pathway claims against cited evidence, validates required fields and workload limits, and either accepts the draft or flags it for

Metric Category	Your Response
e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.	repair. SDG links are generated by matching ESCO occupations and skills against the UN's 169 targets; uncertain role matches fall back to SDG 8.5 and are labelled as a baseline. For the secondary rising-skill KPI, the held-out O*NET evaluation reports **1.39× lift over a 2.1% base rate**. This is a historical model result, not a measured change in employment or learner outcomes. The previous phase target was **≥95% of activities passing the evidence check on the first draft**, with **100% of activities shown for approval verified or flagged**. The folder does not contain an aggregate run-time measurement of those percentages or a user pilot, so those targets are not yet demonstrated as achieved. They should be measured from Reviewer outcomes on a defined set of generated plans.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	AiversePulse is designed to help workers, graduates and career switchers identify their skill gaps and organise learning around their available time. Its scenarios comparison helps users explore how different assumptions about AI adoption affect their learning priorities. University career centres, TVET providers and employer learning-and-development teams are potential delivery partners. They could use AiversePulse to review recommendations and help users select suitable learning resources. For organizations, the platform can also support more targeted workforce-development decisions by identifying skill gaps and prioritising relevant training. The platform supports human decision-making through reviewable plans rather than treating AI-generated recommendations as guaranteed career outcomes. A future pilot would measure the pathway usefulness, learning completion and changes in assessed skills. Longer-term employment and income effects would require follow-up evaluation.

Metric Category	Your Response
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	AiversePulse uses a browser frontend and a FastAPI backend with streamed workflow updates. Its operational database layer supports SQLite and configurable PostgreSQL. The workflow reuses unaffected outputs when users change their learning hours, skills or scenario, reducing unnecessary processing. Initial adoption could begin through a supervised institutional or employer pilot. Pilot evaluation would measure reliability, cost per generated plan and user usefulness. Wider deployment would depend on acceptable operating costs, source freshness and appropriate handling of personal data.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

AiversePulse uses deterministic validation to make generated reskilling plans traceable and feasible. The Reviewer checks that cited evidence IDs belong to the retrieved evidence set, activities address documented skill gaps, estimated hours are positive, and each phase fits the learner's available time. Failed checks trigger up to two targeted repair attempts. Unresolved errors result in a flagged plan requiring human attention.

These controls support SDG 8.5 and SDG 4.4 by helping learners receive structured, reviewable recommendations within their available learning time.

A. Learning Time Calculation

The available learning time for each 30-day phase is calculated as:

Phase capacity = Weekly learning hours × 30 ÷ 7

The result is rounded to one decimal place. Each phase must satisfy:

Total estimated activity hours ≤ Phase capacity

For a learner with six available hours per week:

Phase capacity = 6 × 30 ÷ 7 = **25.7 hours**

Across three phases, the calculated scheduling capacity is **77.1 hours**. Skills whose estimated learning requirements exceed the available capacity are deferred.

The following excerpts show the implemented calculation and capacity check:

```python
DAYS_PER_PHASE = 30
WEEKS_PER_PHASE = DAYS_PER_PHASE / 7


Windsurf: Refactor | Explain | Generate Docstring | ×
def phase_capacity(weekly_hours: float) -> float:
    return round(weekly_hours * DAYS_PER_PHASE / 7, 1)
```

```python
def validate(p: Pathway, gaps: list[Gap], evidence_ids: set[str], weekly_hours: float) -> list[ValidationIssue]:
    from agents.pathway import PHASES, phase_capacity
    issues = []
    gap_skills = {g.skill_id for g in gaps if g.gap > 0}
    cap = phase_capacity(weekly_hours)
    for i, a in enumerate(p.activities):
        f = f"activities[{i}]"
        if a.estimated_hours <= 0:
            issues.append(ValidationIssue(code="bad_hours", severity="error", field=f, message="non-positive hours"))
        if a.skill_id not in gap_skills:
            issues.append(ValidationIssue(code="no_gap", severity="error", field=f,
                                          message=f"{a.skill_id} has no documented gap"))
        if bad := [e for e in a.evidence_ids if e not in evidence_ids]:
            issues.append(ValidationIssue(code="unknown_evidence", severity="error", field=f,
                                          message=f"citation not retrieved: {bad}"))
    for ph in PHASES:
        h = sum(a.estimated_hours for a in p.activities if a.day_range == ph)
        if h > cap + 1e-6:
            issues.append(ValidationIssue(code="over_capacity", severity="error", field=f"phase {ph}",
                                          message=f"{h:.1f}h planned > {cap:.1f}h available"))
```

B. Bounded Repair and Human Review

The Reviewer permits a maximum of two repair attempts. After each attempt, the plan is validated again. A plan with unresolved errors receives the status “flagged”; a plan without detected errors remains a “draft” for human review.

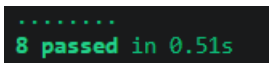
```python
def revise(p: Pathway, gaps, evidence_ids, weekly_hours, repair_fn=repair):
    """-> (pathway with status draft|flagged, issues, revision_count)."""
    issues = validate(p, gaps, evidence_ids, weekly_hours)
    n = 0
    while any(i.severity == "error" for i in issues) and n < MAX_REVISIONS:
        p = repair_fn(p, issues, evidence_ids, weekly_hours)
        n += 1
        issues = validate(p, gaps, evidence_ids, weekly_hours)
    status = "flagged" if any(i.severity == "error" for i in issues) else "draft"
    return p.model_copy(update={"status": status}), issues, n

```

C. Verification Results

On 30 October 2026, the existing revision and approval tests were executed locally using:

Command: .\.*venv\Scripts\python.exe -m pytest tests/test_revision.py tests/test_approval.py -q*
Result:



5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

With corporate funding or institutional partnerships, we would first validate AiversePulse through a supervised pilot with a university, employer or civic training partner.

The pilot would measure first-draft evidence compliance, reviewer corrections, learning completion, changes in assessed skills, user satisfaction and cost per completed plan. Longer follow-up would examine career transitions and employment outcomes. Evaluation would distinguish real participant records from demonstration data.

For higher volumes, we would build on the existing PostgreSQL adapter and benchmark and optimise the existing PostgreSQL and pgvector configuration for larger workloads. We would add a durable background-job queue, shared rate limiting and stronger monitoring. Cached occupational knowledge and model outputs would reduce repeated computation. Model routing would assign smaller models to simpler tasks and reserve more capable models for complex generation.

Partner integrations would prioritise HR-system imports, learning-management systems and verified course catalogues. Local employer input would help adapt occupational requirements to Malaysian roles, while multilingual interfaces would broaden access.

The commercial model would combine individual subscriptions; organisation plans and institution-funded learner access. Funding would initially support evaluation, reliable hosting, privacy controls, accessibility and partner onboarding.

We would expand only after pilot results demonstrate useful learning outcomes and acceptable operating costs. Potential Rotary and XMUM partnerships could provide pilot participants, mentors and independent evaluation.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET 31.0	dataset	CC BY 4.0 (USDOL/ETA)
sentence-transformers/all-MiniLM-L6-v2	model	Apache-2.0
FastMCP	code	Apache-2.0
pdf-prompt-injection-toolkit	code	MIT
ESCO v1.2 occupations and skills (European Commission)	dataset	EC reuse policy, attribution required
UN SDG goals and targets (UN Statistics SDG API)	dataset	© United Nations; UN terms of use
LangGraph;	code	MIT;
FastAPI;	code	MIT;

PyTorch;	code	BSD-3-Clause;
ChromaDB	code	Apache-2.0
Gemini 3.1 Flash-Lite; gateway models (grok-4.7, gpt-5.6-luna, mimo-v2.6-flash)	model (API)	Provider API terms
Three.js; GSAP / ScrollTrigger	code	MIT; GSAP Standard License

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

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8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



[Signature]

30/10/2026

[Date]